

# A comparison of three highly active antiretroviral treatment strategies consisting of non-nucleoside reverse transcriptase inhibitors, protease inhibitors, or both in the presence of nucleoside reverse transcriptase inhibitors as initial therapy (CPCRA 058 FIRST Study): a long-term randomised trial

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## Summary

**Background** Long-term data from randomised trials on the consequences of treatment with a protease inhibitor (PI), non-nucleoside reverse transcriptase inhibitor (NNRTI), or both are lacking. Here, we report results from the FIRST trial, which compared initial treatment strategies for clinical, immunological, and virological outcomes.

**Methods** Between 1999 and 2002, 1397 antiretroviral-treatment-naïve patients, presenting at 18 clinical trial units with 80 research sites in the USA, were randomly assigned in a ratio of 1:1:1 to a protease inhibitor (PI) strategy (PI plus nucleoside reverse transcriptase inhibitor [NRTI]; n=470), a non-nucleoside reverse transcriptase inhibitor (NNRTI) strategy (NNRTI plus NRTI; n=463), or a three-class strategy (PI plus NNRTI plus NRTI; n=464). Primary endpoints were a composite of an AIDS-defining event, death, or CD4 cell count decline to less than 200 cells per mm<sup>3</sup> for the PI versus NNRTI comparison, and average change in CD4 cell count at or after 32 months for the three-class versus combined two-class comparison. Analyses were by intention-to-treat. This study is registered with ClinicalTrials.gov, number NCT00000922.

**Findings** 1397 patients were assessed for the composite endpoint. A total of 388 participants developed the composite endpoint, 302 developed AIDS or died, and 188 died. NNRTI versus PI hazard ratios (HRs) for the composite endpoint, for AIDS or death, for death, and for virological failure were 1.02 (95% CI 0.79–1.31), 1.07 (0.80–1.41), 0.95 (0.66–1.37), and 0.66 (0.56–0.78), respectively. 1196 patients were assessed for the three-class versus combined two-class primary endpoint. Mean change in CD4 cell count at or after 32 months was +234 cells per mm<sup>3</sup> and +227 cells per mm<sup>3</sup> for the three-class and the combined two-class strategies (p=0.62), respectively. HRs (three-class vs combined two-class) for AIDS or death and virological failure were 1.15 (0.91–1.45) and 0.87 (0.75–1.00), respectively. HRs (three-class vs combined two-class) for AIDS or death were similar for participants with baseline CD4 cell counts of 200 cells per mm<sup>3</sup> or less and of more than 200 cells per mm<sup>3</sup> (p=0.38 for interaction), and for participants with baseline HIV RNA concentrations less than 100 000 copies per mL and 100 000 copies per mL or more (p=0.26 for interaction). Participants assigned the three-class strategy were significantly more likely to discontinue treatment because of toxic effects than were those assigned to the two-class strategies (HR 1.58; p<0.0001).

**Interpretation** Initial treatment with either an NNRTI-based regimen or a PI-based regimen, but not both together, is a good strategy for long-term antiretroviral management in treatment-naïve patients with HIV.

## Introduction

Highly active antiretroviral therapy (HAART) has been available for the treatment of HIV-infected individuals for more than a decade. However, during this time there have been no long-term randomised clinical trials to compare different initial antiretroviral strategies for clinical outcomes (HIV disease progression to AIDS or death). Previous short-term studies that compared either different drugs within a class or a protease inhibitor (PI)-based regimen with a non-nucleoside reverse transcriptase

inhibitor (NNRTI)-based regimen and that found differences in HIV RNA responses between the groups were not designed to detect differences in clinical outcome.<sup>1,2</sup> The best clinical event data currently available are based on observational data, such as the EuroSIDA cohort of 6814 people with 22767 person-years of follow up.<sup>3</sup> The EuroSIDA study showed that the rate of AIDS or death did not differ for different combinations of antiretroviral drugs in people with similar CD4-cell counts and HIV RNA concentrations.

*Lancet* 2006; 368: 2125–35

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The FIRST (Flexible Initial Retrovirus Suppressive Therapies) trial was one of three large long-term randomised clinical trials started between 1998 and 1999 to assess the efficacy and safety of initial antiretroviral treatment strategies.<sup>4-6</sup> In all three trials, triple-class treatment, consisting of PIs, an NNRTI, plus nucleoside reverse transcriptase inhibitors (NRTIs), was compared with both a PI-based strategy and an NNRTI-based strategy. The other two trials (ACTG 384 and INITIO)<sup>4,5</sup> examined prespecified antiretroviral drugs for the initial and second rounds of treatment over a shorter follow-up period than in the FIRST trial and for virological and immunological outcomes. The FIRST trial was designed with a flexible framework: the choice of specific drugs within a class was made by the treating clinician in partnership with the participant. Here, we report results from the FIRST trial, which compared the three initial treatment strategies for clinical, immunological, and virological outcomes to answer two questions: 1) is it better to begin initial antiretroviral treatment with a three-class strategy than with a two-class strategy? and 2) which of the two-class strategies, PI or NNRTI, is better as initial therapy? The results of the FIRST study provide important insight into how the three strategies compare for long-term clinical outcomes as well as immunological and virological endpoints.

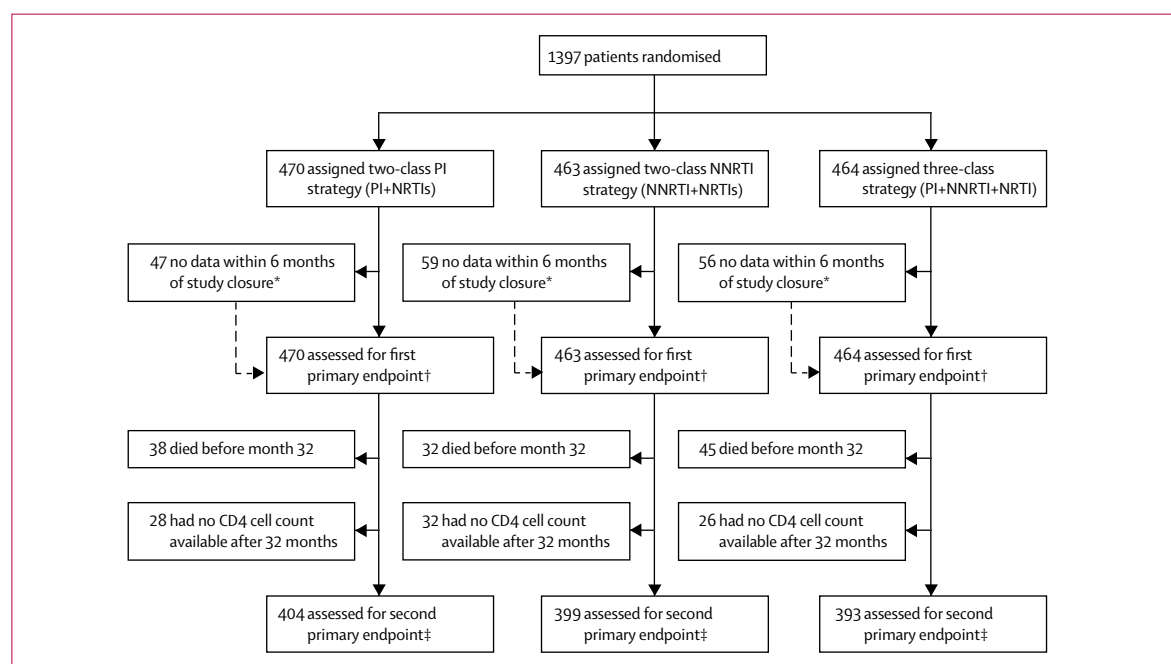
## Methods

### Study population and design

The FIRST study was undertaken by the Community Programs for Clinical Research on AIDS (CPCRA)

network, which consisted of 18 clinical trial units with 80 research sites in the USA. The institutional review board of each unit approved the protocol and all participants or guardians gave written informed consent. Individuals with HIV were eligible for inclusion if they were at least 13 years of age, had not previously used a PI or an NNRTI, had not used NRTIs for a cumulative time of more than 4 weeks, and had not used lamivudine for more than 1 week. Eligible patients were randomly allocated, in a ratio of 1:1:1, to one of three starting strategies: PI strategy (PI+NRTI); NNRTI strategy (NNRTI+NRTI); or three-class strategy (PI+NNRTI+NRTI). Randomisation was stratified by clinical unit and CD4 cell count ( $\leq 200$  vs  $>200$  cells per  $\text{mm}^3$ ). Randomisation schedules for each stratum were generated with randomly mixed permuted blocks of size three and six. The randomisation schedules were generated centrally by the CPCRA Statistical Centre randomisation committee. Members of that committee had no other involvement with the FIRST study. Assignments were obtained by calling a toll-free number at the CPCRA statistical centre.

Before randomisation, participants and investigators could either preselect the specific drugs used or consent to an additional randomisation of drugs within each class.<sup>6,7</sup> For the two strategies that use two classes of drugs two NRTIs were to be used; for the three-class strategy one or two NRTIs could be used. For the two strategies containing PIs, one or two PIs could be used. Strategy and regimen changes were allowed at any time. When possible, guidelines prepared and periodically



**Figure 1: Study design and description of cohort available for endpoint determination**

\*Participants were included in all analyses using all available data. †AIDS defining event, death, or CD4 cell count decline to  $<200$  cells per  $\text{mm}^3$ . ‡Average change in CD4 cell count beginning with month 32.

|                                                     | PI strategy (PI+NRTI)<br>(n=470) | NNRTI strategy<br>(NNRTI+NRTI) (n=463) | Combined two-class<br>strategies (n=933) | Three-class strategy<br>(PI+NNRTI+NRTI) (n=464) | Total (n=1397)   |
|-----------------------------------------------------|----------------------------------|----------------------------------------|------------------------------------------|-------------------------------------------------|------------------|
| <b>Baseline characteristics</b>                     |                                  |                                        |                                          |                                                 |                  |
| Median age, years (IQR)                             | 38 (31–44)                       | 38 (32–43)                             | 38 (32–44)                               | 38 (31–44)                                      | 38 (32–44)       |
| Women                                               | 107 (23%)                        | 96 (21%)                               | 203 (22%)                                | 85 (18%)                                        | 288 (21%)        |
| Ethnic origin                                       |                                  |                                        |                                          |                                                 |                  |
| Black                                               | 255 (54%)                        | 249 (54%)                              | 504 (54%)                                | 247 (53%)                                       | 751 (54%)        |
| Latino                                              | 83 (18%)                         | 77 (17%)                               | 160 (17%)                                | 78 (17%)                                        | 238 (17%)        |
| White                                               | 114 (24%)                        | 127 (27%)                              | 241 (26%)                                | 127 (27%)                                       | 368 (26%)        |
| Other                                               | 18 (4%)                          | 10 (2%)                                | 28 (3%)                                  | 12 (3%)                                         | 40 (3%)          |
| Intravenous drug use (%)                            | 72 (15%)                         | 70 (15%)                               | 142 (15%)                                | 68 (15%)                                        | 210 (15%)        |
| Prior AIDS-defining event (%)                       | 184 (39%)                        | 179 (39%)                              | 363 (39%)                                | 164 (35%)                                       | 527 (38%)        |
| Median CD4 cell count (cells per mm <sup>3</sup> )* | 162.8                            | 169.5                                  | 164.5                                    | 149.3                                           | 162.5            |
| IQR                                                 | (44.5–329)                       | (36.0–329.0)                           | (39.0–329.0)                             | (31.8–337.0)                                    | (36.0–331.5)     |
| Median HIV RNA concentration (copies per mL)*       | 148 195                          | 148 947                                | 148 947                                  | 126 361                                         | 143 844          |
| IQR                                                 | (38 700–424 297)                 | (34 017–430 935)                       | (37 110–428 330)                         | (41 402–349 540)                                | (38 418–400 147) |
| <b>Antiretroviral treatment prescribed (%)</b>      |                                  |                                        |                                          |                                                 |                  |
| <b>PI</b>                                           |                                  |                                        |                                          |                                                 |                  |
| Nelfinavir                                          | 285 (61%)                        | 3 (0.6%)                               | ..                                       | 292 (63%)                                       |                  |
| Indinavir                                           | 53 (11%)                         | 0                                      | ..                                       | 53 (11%)                                        |                  |
| Ritonavir-boosted PI                                |                                  |                                        | ..                                       |                                                 |                  |
| Indinavir                                           | 65 (14%)                         | 0                                      | ..                                       | 57 (12%)                                        |                  |
| Lopinavir                                           | 37 (8%)                          | 0                                      | ..                                       | 32 (7%)                                         |                  |
| Other ritonavir-boosted PI                          | 20 (4%)                          | 0                                      | ..                                       | 16 (3%)                                         |                  |
| Other PI                                            | 8 (2%)                           | 0                                      | ..                                       | 11 (2%)                                         |                  |
| No PI                                               | 2 (0.4%)                         | 460 (99%)                              | ..                                       | 3 (1%)                                          |                  |
| <b>NNRTI</b>                                        |                                  |                                        |                                          |                                                 |                  |
| Efavirenz                                           | 0                                | 293 (63%)                              | ..                                       | 242 (52%)                                       |                  |
| Nevirapine                                          | 2 (0.4%)                         | 169 (37%)                              | ..                                       | 211 (46%)                                       |                  |
| Other NNRTI                                         | 0                                | 1 (0.2%)                               | ..                                       | 10 (2%)                                         |                  |
| No NNRTI                                            | 468 (100%)                       | 0                                      | ..                                       | 1 (0.2%)                                        |                  |
| <b>NRTI</b>                                         |                                  |                                        |                                          |                                                 |                  |
| Zidovudine+lamivudine                               | 261 (56%)                        | 266 (58%)                              | 527 (57%)                                | 208 (45%)                                       |                  |
| Stavudine+didanosine                                | 43 (9%)                          | 39 (8%)                                | 82 (9%)                                  | 39 (8%)                                         |                  |
| Abacavir+lamivudine                                 | 42 (9%)                          | 40 (9%)                                | 82 (9%)                                  | 43 (9%)                                         |                  |
| Stavudine+abacavir                                  | 18 (4%)                          | 12 (3%)                                | 30 (3%)                                  | 4 (1%)                                          |                  |
| Stavudine+lamivudine                                | 99 (21%)                         | 94 (20%)                               | 193 (21%)                                | 51 (11%)                                        |                  |
| Single NRTI                                         | 1 (0.2%)                         | 1 (0.2%)                               | 2 (0.2%)                                 | 111 (24%)                                       |                  |
| Other NRTIs                                         | 6 (1%)                           | 11 (2%)                                | 17 (2%)                                  | 7 (2%)                                          |                  |
| No NRTI                                             | 0                                | 0                                      | 0                                        | 1 (0.2%)                                        |                  |

\*Average of screening and baseline concentrations.

**Table 1: Baseline characteristics and antiretroviral treatment prescribed after randomisation**

updated by the US Department of Health and Human Services<sup>8</sup> were to be followed for changing antiretroviral treatment for drug failure. As per the guidelines, the goal of treatment was to maintain virological suppression with the recognition that many factors (such as poor adherence to the current regimen) could have an effect on the decision to change or continue a participant's regimen. When drugs were discontinued because of toxic effects, drugs within the same class were to be substituted where possible to maintain patients on their randomly assigned strategy.

For comparison of the PI and NNRTI strategies, the primary endpoint was a composite of an AIDS-defining event, death, or CD4 cell count decline to less than 200 cells per mm<sup>3</sup> (assessed only in those with baseline CD4 cell counts  $\geq 200$  cells per mm<sup>3</sup>). The clinical and CD4 cell count endpoints were thought to be more appropriate outcomes for a long-term trial than were virological endpoints because CD4 cell count is a better predictor of risk of disease progression than HIV RNA concentration in people on antiretroviral treatment.<sup>9</sup> For this comparison, 292 primary endpoints were

|                                                                             | PI strategy<br>(PI+NRTI)<br>(n=470) | NNRTI strategy<br>(NNRTI+NRTI)<br>(n=463) | Three-class strategy<br>(PI+NNRTI+NRTI)<br>(n=464) |
|-----------------------------------------------------------------------------|-------------------------------------|-------------------------------------------|----------------------------------------------------|
| Switched* strategy at least once                                            | 203 (43%)                           | 149 (32%)                                 | 369 (80%)                                          |
| Median time to first switch in strategy†, months (IQR)                      | 18.2 (8.5–32.7)                     | 23.5 (11.1–40.3)                          | 8.8 (1.4–24.1)                                     |
| Reasons‡ for first switch in strategy                                       |                                     |                                           |                                                    |
| Toxic effects                                                               | 41 (9%)                             | 20 (4%)                                   | 136 (29%)                                          |
| Viral burden                                                                | 57 (12%)                            | 59 (13%)                                  | 28 (6%)                                            |
| Other                                                                       | 143 (30%)                           | 95 (21%)                                  | 256 (55%)                                          |
| Percentage of follow-up time                                                |                                     |                                           |                                                    |
| On PI                                                                       | 63.9%                               | 13.5%                                     | 21.3%                                              |
| On NNRTI                                                                    | 15.4%                               | 69.9%                                     | 15.5%                                              |
| On PI+NNRTI                                                                 | 2.8%                                | 0.9%                                      | 45.8%                                              |
| On NRTI only                                                                | 4.7%                                | 2.8%                                      | 3.7%                                               |
| No antiretroviral therapy                                                   | 13.3%                               | 12.9%                                     | 13.7%                                              |
| At least one antiretroviral treatment change                                | 410 (87%)                           | 331 (72%)                                 | 407 (88%)                                          |
| Discontinued all antiretroviral treatment at least once for 30 days or more | 208 (44%)                           | 201 (44%)                                 | 204 (44%)                                          |
| Median antiretroviral exposure, months (IQR)                                |                                     |                                           |                                                    |
| PI drugs                                                                    | 35.3 (16.7–52.8)                    | 0 (0–11.6)                                | 39.0 (13.6–55.1)                                   |
| NNRTI drugs                                                                 | 0 (0–16.8)                          | 41.0 (17.3–57.6)                          | 29.3 (10.7–53.9)                                   |
| NRTI drugs                                                                  | 50.0 (36.2–62.4)                    | 51.9 (35.1–62.0)                          | 50.6 (32.3–61.2)                                   |
| Number of antiretroviral drugs ever exposed to§                             |                                     |                                           |                                                    |
| Three drugs                                                                 | 95 (20%)                            | 208 (45%)                                 | 16 (3%)                                            |
| Four or five drugs                                                          | 137 (29%)                           | 98 (21%)                                  | 196 (42%)                                          |
| Six or seven drugs                                                          | 111 (24%)                           | 73 (16%)                                  | 122 (26%)                                          |
| Eight or more drugs                                                         | 127 (27%)                           | 83 (18%)                                  | 128 (28%)                                          |

\*A switch in strategy is defined as adding an NNRTI for the PI strategy, adding a PI for the NNRTI strategy, dropping all the PIs or the NNRTI or both in the three-class strategy. †Among those who switched strategies. ‡Reasons are not mutually exclusive. §One patient in the NNRTI strategy was exposed to two drugs and two patients in the three-class strategy were not exposed to any drugs.

**Table 2: Adherence to treatment strategies during follow-up**

planned for 80% power to detect a hazard ratio of 0.67. For the comparison of the three-class strategy with the pooled results of the two-class strategies, the primary endpoint was the average change in CD4 cell count beginning with month 32. For this comparison, total sample size needed to detect a 40-cell difference between the groups with 80% power was 1176. A significance level of 0.05 (two-sided) was used for both comparisons. Secondary endpoints analysed in this report include death, an AIDS-defining event or death, change of the initial treatment strategy, virological suppression, drug resistance at virological failure, toxic effects and grade 4 adverse events, and self-reported adherence to antiretroviral treatment.

#### Data collection and monitoring

After randomisation, participants had study visits at months 1 and 4 and then every 4 months thereafter. At each visit, clinical status was assessed and information was obtained about changes in antiretroviral treatment for 30 days or more and self-reported adherence based on antiretroviral drugs taken in the previous 7 days. CD4 cell

count, HIV RNA concentration (Roche Amplicor 1.0), fasting or non-fasting blood lipids, glucose, aspartate transferase, and alanine transferase were measured at each visit. Disease progression events were defined as the occurrence of a confirmed or probable AIDS-defining event according to modified US Centers for Disease Control (CDC) criteria (1993).<sup>10</sup> A clinical events committee unaware of the assigned treatment group reviewed all disease progression events and deaths. All grade 4 adverse events were reported irrespective of the presumed association with antiretroviral treatment. Lower grade adverse events (grades 1–3) that led to discontinuation of antiretroviral treatment also were reported. Adverse events were coded according to a toxicity manual developed by the National Institute of Allergy and Infectious Diseases, Division of AIDS.<sup>11</sup> A switch in strategy was defined as addition of an NNRTI in the PI strategy, addition of a PI in the NNRTI strategy, and removal of all PIs, the NNRTI, or both in the three-class strategy.

Viral rebound was defined as an HIV RNA concentration of more than 400 copies per mL, on the condition of having first achieved an HIV RNA concentration of less than 50 copies per mL and having at least one viral load measurement after that point. This was not a predefined endpoint but was added for comparison with other major trials. Virological failure was defined as an HIV RNA concentration of more than 1000 copies per mL at or after the 4 month visit. Genotypic resistance testing was undertaken at virological failure. Resistance was defined as definite drug resistance by genotype (TRUGENE HIV-1 Genotyping Kit and CPCRA interpretive algorithm v4.0). The presence of specific drug-resistance mutations was assessed by use of the October, 2004, International AIDS Society, USA (IAS-USA) tables.<sup>12</sup>

During the study all investigators were blinded toward the study data gathering. An independent data and safety monitoring board reviewed interim analyses at least once per year. The O'Brien-Fleming boundaries and the Lan-DeMets spending function were used to monitor the primary event comparisons at the six interim analyses.

#### Statistical analysis

Unless otherwise stated all analyses were by intention to treat, with all participants assessed in the strategy groups to which they were originally randomised even if no treatment was started after randomisation. Analyses of the primary endpoints were stratified or adjusted for clinical unit and baseline CD4 cell count, consistent with randomisation. Time to event methods (Kaplan-Meier survival curves and Cox's proportional hazards models) were used to compare groups for: the composite endpoint of an AIDS-defining event, death, or CD4 cell count decline to less than 200 cells per mm<sup>3</sup>; the components of the composite event; grade 4 adverse events; switch in treatment strategy; HIV RNA concentration less than 50 copies per mL; viral rebound; and virological failure

with and without drug resistance. The proportional hazards assumption was tested by including an interaction between treatment group and log-transformed follow-up time. The censoring date used in clinical event analyses was the common closing date for the study (Sept 16, 2005) or the date the patient was lost to follow-up for the event assessed. One site with 30 patients closed during the study on Feb 22, 2004, for administrative reasons; follow-up of patients at that site is censored on that closing date. All available measurements through the common closing date were used for analyses that included CD4 cell count or HIV RNA concentration. Hazard ratios (HR) are given with 95% CIs. All rates are per 100 person-years of follow-up; p values corresponding to a 2-degree of freedom omnibus test for comparing the three groups also are cited. For sensitivity analyses, selected on-treatment analyses were undertaken.

Analysis of variance stratified by clinical unit was used to compare the combined two-class and three-class strategy groups for change in CD4 cell count from baseline, averaging over all CD4 cell counts at or after 32 months and adjusting for baseline CD4 cell count. Longitudinal regression models<sup>13</sup> are used to compare groups for changes in CD4 cell counts according to all follow-up measurements available and to compare rate of change in CD4 cell count beginning at month 8. Longitudinal regression models for binary data were used to estimate the odds of having achieved a HIV RNA concentration of less than 50 copies per mL during follow-up. Differences in proportions were compared using  $\chi^2$  tests.

Major outcomes are summarised for predefined baseline subgroups: age, sex, ethnic origin, history of injecting drug use, a previous AIDS-defining event and CD4 cell count, and HIV RNA concentrations. Lack of homogeneity of treatment differences for subgroups was assessed by including treatment by subgroup interaction terms in Cox models and regression models; p values

corresponding to these tests of homogeneity are cited for each subgroup. Statistical analyses were undertaken with SAS (version 8.2).

This study is registered with ClinicalTrials.gov, number NCT00000922.

### Role of the funding source

The sponsor of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report, except through membership with the protocol team. After study closure, the corresponding author had full access to all the data in the study and had final responsibility for the decision to submit for publication.

### Results

Between February, 1999, and January, 2002, 1397 individuals were randomly assigned to a treatment group: 470 to the PI strategy, 463 to the NNRTI strategy, and 464 to the three-class strategy (figure 1). The three strategy groups were well balanced in terms of demographic and other baseline characteristics (table 1). Most patients met the current US Department of Health and Human Services (DHHS) guidelines for antiretroviral treatment, with 62% reporting a previous AIDS-defining event or a CD4 cell count of 200 cells per mm<sup>3</sup> or less (18% had a CD4 count 201–350 cells per mm<sup>3</sup> and 20% had a CD4 count >350 cells per mm<sup>3</sup>). Around 6% and 20% of participants were co-infected with hepatitis B (hepatitis B surface antigen positive) and hepatitis C (hepatitis C antibody positive), respectively.

Nelfinavir was the most commonly used PI in patients assigned to the PI strategy, followed by a ritonavir-boosted PI (table 1). The proportion of participants prescribed a ritonavir-boosted PI changed over the 3 years of enrolment ( $p<0.0001$ ). In the first year of enrolment, 15% of all participants were prescribed a ritonavir-boosted PI, whereas in the second and third years of enrolment

|                                                                                                                                | Number of events and rate per 100 person years |      |                                     |      |                                       |      |                                              |      | Hazard ratio (95% CI) |                          | p for difference among the three randomised groups |
|--------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|------|-------------------------------------|------|---------------------------------------|------|----------------------------------------------|------|-----------------------|--------------------------|----------------------------------------------------|
|                                                                                                                                | PI strategy (PI+NRTI) (n=470)                  |      | NNRTI strategy (NNRTI+NRTI) (n=463) |      | Combined Two-class strategies (n=933) |      | Three-class strategy (PI+NNRTI+NRTI) (n=464) |      | NNRTI vs PI           | Three-class vs two-class |                                                    |
|                                                                                                                                | N                                              | Rate | N                                   | Rate | N                                     | Rate | N                                            | Rate |                       |                          |                                                    |
| AIDS, death, or CD4 cell count <200 cells per mm <sup>3</sup> *†                                                               | 126                                            | 6.6  | 122                                 | 6.7  | 248                                   | 6.6  | 140                                          | 7.8  | 1.02 (0.79–1.31)      | 1.15 (0.94–1.42)         | 0.40                                               |
| Death                                                                                                                          | 60                                             | 2.8  | 56                                  | 2.6  | 116                                   | 2.7  | 72                                           | 3.4  | 0.95 (0.66–1.37)      | 1.28 (0.95–1.71)         | 0.26                                               |
| AIDS                                                                                                                           | 57                                             | 2.9  | 67                                  | 3.5  | 124                                   | 3.2  | 60                                           | 3.2  | 1.21 (0.85–1.72)      | 0.98 (0.72–1.33)         | 0.56                                               |
| AIDS or death                                                                                                                  | 95                                             | 4.8  | 98                                  | 5.1  | 193                                   | 5.0  | 109                                          | 5.7  | 1.07 (0.80–1.41)      | 1.15 (0.91–1.45)         | 0.48                                               |
| CD4 cell count <200 cells per mm <sup>3</sup> †                                                                                | 41                                             | 4.9  | 32                                  | 3.9  | 73                                    | 4.4  | 38                                           | 4.8  | 0.81 (0.51–1.29)      | 1.08 (0.73–1.60)         | 0.61                                               |
| Grade 4 event                                                                                                                  | 147                                            | 8.7  | 147                                 | 8.9  | 294                                   | 8.8  | 145                                          | 8.9  | 1.02 (0.81–1.28)      | 1.01 (0.83–1.23)         | 0.98                                               |
| AIDS, death, or grade 4 event                                                                                                  | 187                                            | 11.5 | 191                                 | 12.4 | 378                                   | 12.0 | 192                                          | 12.5 | 1.06 (0.87–1.30)      | 1.04 (0.88–1.24)         | 0.76                                               |
| *Primary endpoint for the NNRTI versus PI comparison. †For those with baseline CD4 cell count ≥200 cells per mm <sup>3</sup> . |                                                |      |                                     |      |                                       |      |                                              |      |                       |                          |                                                    |
| Table 3: Major endpoints by treatment strategy                                                                                 |                                                |      |                                     |      |                                       |      |                                              |      |                       |                          |                                                    |

Table 3: Major endpoints by treatment strategy

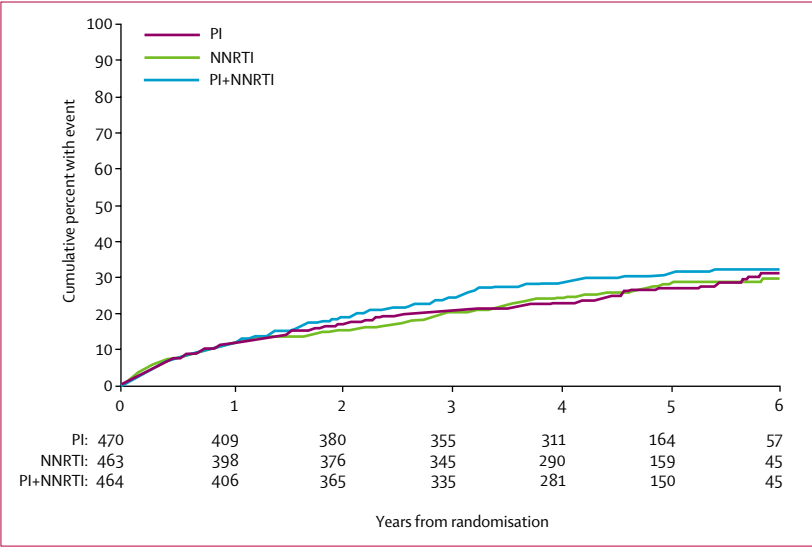


Figure 2: Kaplan-Meier estimates of time to the composite endpoint of an AIDS-defining event, death, or CD4 cell-count decline to less than 200 cells per mm<sup>3</sup> by treatment strategy

28% and 32% were prescribed a ritonavir-boosted PI, respectively. Among those assigned the NNRTI strategy, efavirenz was prescribed for nearly two-thirds of patients and nevirapine was prescribed for more than a third. Similar NRTI combinations were used for participants in the PI and NNRTI strategy groups. For the three-class

strategy, nelfinavir and efavirenz were also the most commonly used PI and NNRTI. For all three strategies, the most common NRTI combination was zidovudine plus lamivudine. A single NRTI, usually stavudine, was used by nearly a quarter of those assigned the three-class strategy.

Median follow-up was 60 months (IQR 52–69). In the year before closeout, the proportion of survivors with a CD4 cell count ascertained was 88% for each of the three strategies. The percentage of survivors with at least one CD4 cell count measure at or after month 32 (the primary outcome for the three-class vs the two-class strategy) was 93% for each group (figure 1).

Almost 99% of the initially prescribed regimens followed the randomised treatment strategy definitions (table 1). The initially prescribed regimen was a protocol violation for only five participants in each of the PI and NNRTI strategies and for six participants in the three-class strategy. Around 97% of participants actually started their randomly assigned strategy. There was at least one switch from the randomised treatment strategy during follow-up for 43%, 32%, and 80% of participants in the PI, NNRTI, and three-class strategy groups, respectively (table 2); hazard ratio for time to first switch in strategy was 0.65 (95% CI 0.53–0.81) for the NNRTI versus PI comparison. Among those who switched strategies, the corresponding median times to first switch in strategy were 18 months, 24 months, and

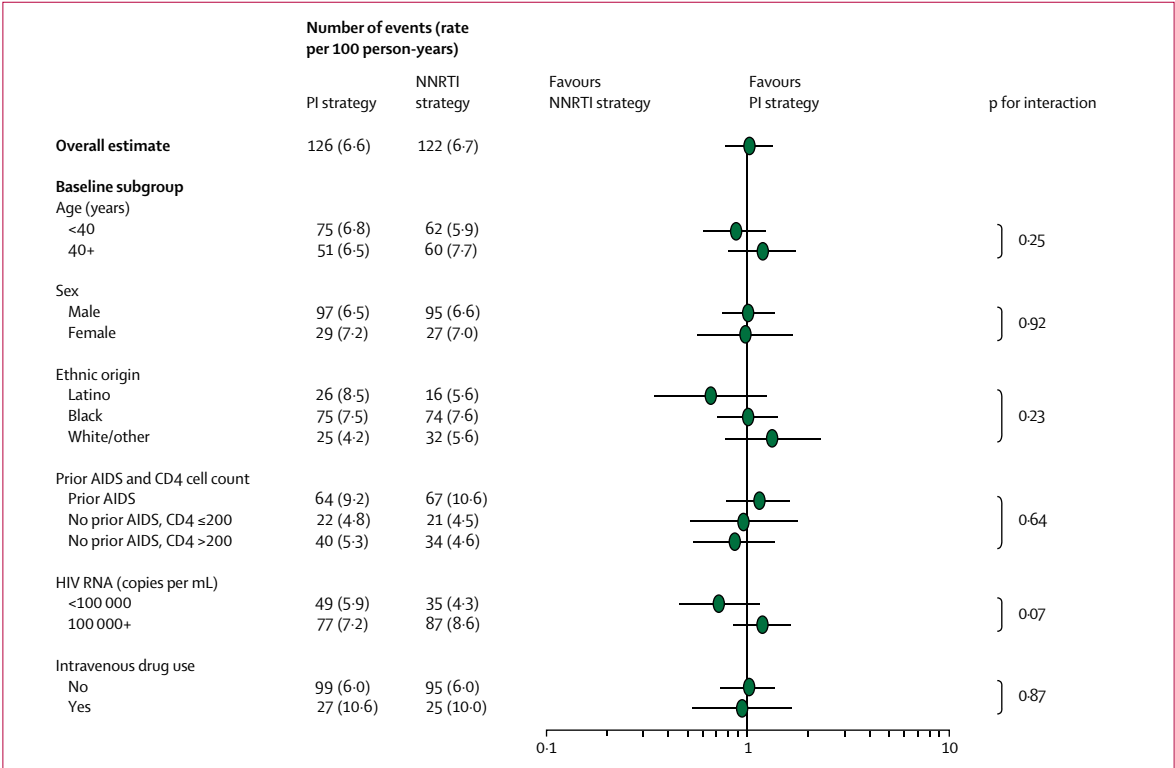
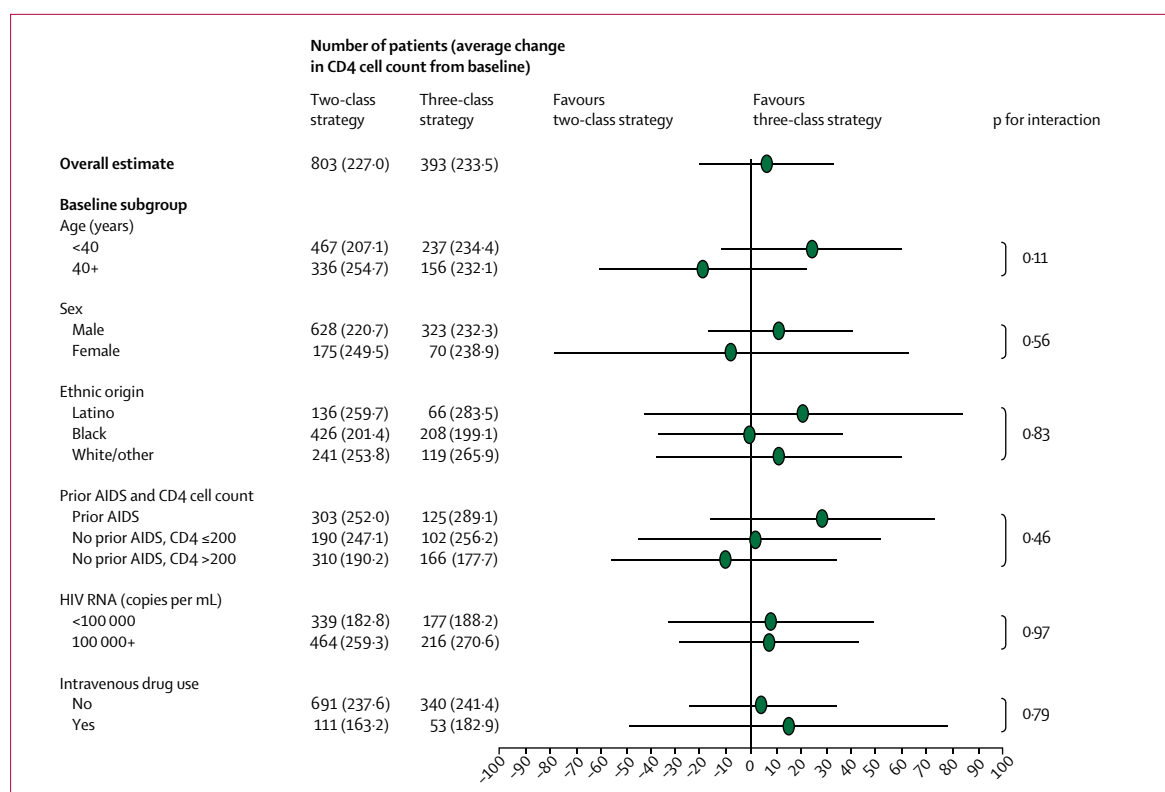


Figure 3: Hazard ratios and 95% CIs for the NNRTI strategy compared to the PI strategy for the composite endpoint of an AIDS-defining event, death, or CD4 cell-count decline to less than 200 cells per mm<sup>3</sup>, overall and for prespecified baseline subgroups





**Figure 4:** Differences and 95% CIs between the three-class strategy and the combined two-class strategies in mean change from baseline in CD4 cell count beginning at month 32, overall and for prespecified baseline subgroups

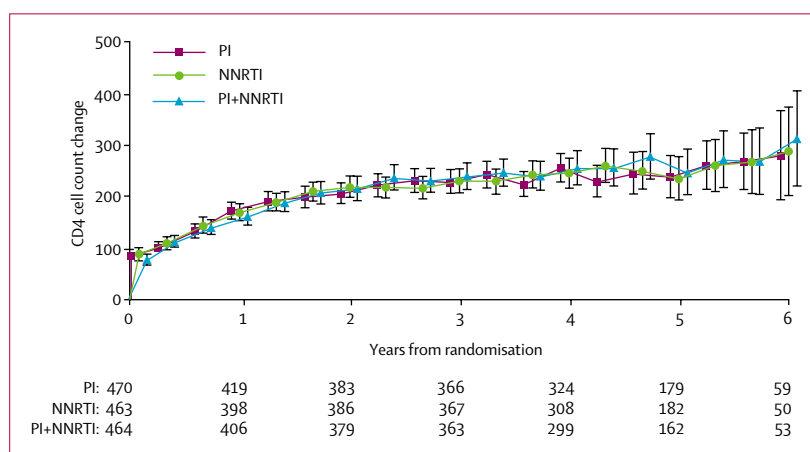
9 months for the three strategies, respectively. The proportion of total follow-up time on the assigned strategy was greatest for the NNRTI strategy and shortest for the three-class strategy.

Because of the greater fraction of follow-up time on assigned strategy, participants assigned to the NNRTI strategy were exposed to fewer antiretroviral drugs than were the other participants. Around 44% of participants in each strategy group discontinued all antiretroviral treatment at least once for 30 days or more (table 2). About 13% of follow-up in all three strategy groups was spent not taking antiretroviral treatment. While taking antiretroviral drugs, self-reported adherence was excellent. At most visits, more than two-thirds of patients prescribed an antiretroviral in each strategy group reported 100% adherence to their regimen. There was no difference in self-reported adherence among the three strategy groups ( $p=0.37$ ).

The rates of discontinuation of one or more drugs in the antiretroviral treatment regimen because of toxic effects were 11.2, 10.3, and 17.6 per 100 person years for the PI, NNRTI, and three-class strategies, respectively (HR 1.58 for the three-class vs two-class comparison,  $p<0.0001$ ). The most common toxic effects that led to treatment discontinuation for the two-class and three-class strategies were: nausea or vomiting (2.6 and 3.1 per 100 person years for two-class and three-class groups,

$p=0.28$ ), diarrhoea (0.9 and 2.2 per 100 person years,  $p<0.0001$ ), and rash (1.1 and 1.8,  $p=0.03$ ). Most (77%) toxic effects that led to discontinuation were grade 1 or grade 2.

For the composite endpoint of an AIDS-defining event, death, or a decline in CD4 cell count to fewer than 200 cells per  $\text{mm}^3$  there was no significant difference between the two-class strategies (table 3, figure 2). The HR estimate did not vary significantly during follow-up (test of



**Figure 5:** Mean change from baseline in CD4 cell count (cells per  $\text{mm}^3$ ) at each follow-up visit by treatment strategy with 95% CIs

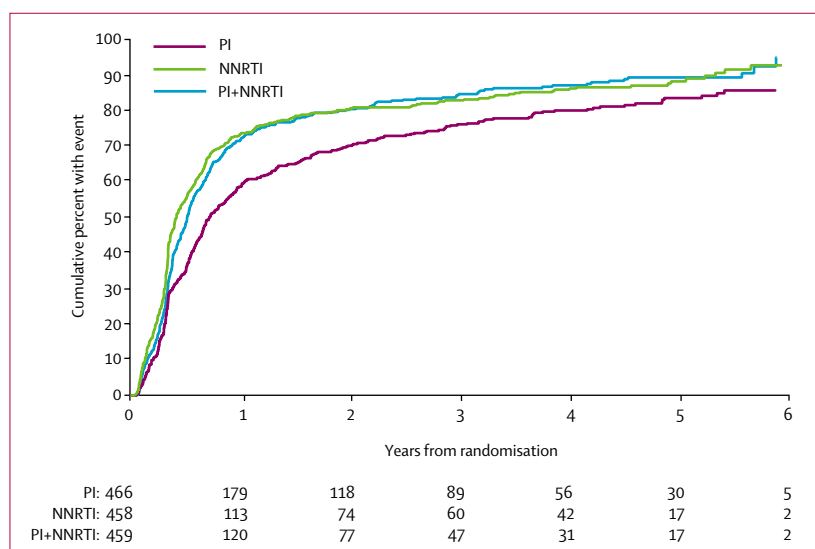


Figure 6: Kaplan-Meier estimates of time to first reduction in HIV RNA concentration to below 50 copies per mL, by treatment strategy

proportional hazards  $p=0.85$ ). The results were consistent with an on-treatment analysis (HR 0.94, 95% CI 0.71–1.25) and across the prespecified baseline subgroups (figure 3). Because the use of ritonavir-boosted PIs changed over time, subgroup analyses by enrolment period were also done for the composite endpoint, with no interaction between strategy group (NNRTI or PI) and year of enrolment ( $p=0.40$  for interaction).

Overall, 188 participants died and 302 had an AIDS-defining event or died (table 3). Neither of these outcomes differed significantly among the three strategies. Hazard ratios comparing the three-class versus combined two-class strategies for an AIDS-defining event or death

were similar for people with baseline CD4 cell counts of less than or equal to 200 cells per  $\text{mm}^3$  and more than 200 cells per  $\text{mm}^3$  ( $p=0.38$  for interaction), and for those with baseline HIV RNA concentrations fewer than 100 000 copies per mL and 100 000 copies per mL or more ( $p=0.26$  for interaction). In all three strategies the rate of an AIDS-defining event or death was greatest in the first 6 months of follow-up (pooled rates were 13.2 per 100 person years vs 3.9 afterwards). Pooled cumulative percentage with an AIDS-defining event or death after 6 months, 12 months, 24 months, 36 months, and 48 months of follow-up were 6%, 9%, 13%, 17%, and 20%, respectively. The most commonly occurring AIDS-defining events were oesophageal candidiasis (53 people), *Pneumocystis jirovecii* pneumonia (36 people), *Mycobacterium avium* complex (25 people), lymphoma (22 people), cryptococcosis (21 people), and cytomegalovirus (20 people).

Average increases in CD4 cell count at or after 32 months of follow-up (the primary endpoint for the three-class vs two-class comparison) were 234 cells per  $\text{mm}^3$  and 227 cells per  $\text{mm}^3$  for the three-class and the combined two-class strategies, respectively (difference=6.7 cells; 95% CI -20 to 33 cells,  $p=0.62$ ). There were no significant differences between the combined two-class and the three-class strategies in average change in CD4 cell count at or after month 32 across the prespecified baseline subgroups (figure 4). Based on longitudinal models that used all available measurements, CD4 cell counts did not differ significantly among the three strategies during follow-up (figure 5;  $p=0.76$ ). The rate of change of CD4 cell count beginning at 8 months of follow-up was similar for the three treatment strategies (+1.7, +1.7, and +1.9 cells per month for the PI, NNRTI, and three-class strategies, respectively;  $p=0.91$ ).

Time to a reduction in HIV RNA concentration to fewer than 50 copies per mL varied significantly among treatment strategies (figure 6;  $p<0.0001$ ). HRs were 1.42 (95% CI 1.23–1.64) for the NNRTI versus PI strategy and 1.13 (1.00–1.28) for the three-class versus combined two-class strategies. HRs for the time for HIV RNA concentration to be reduced to fewer than 50 copies per mL were similar for subgroups defined by year of enrolment ( $p=0.92$  for interaction for the NNRTI versus PI comparison;  $p=0.96$  for the three-class vs two-class comparison). The odds ratios for achieving an HIV RNA concentration of fewer than 50 copies per mL during follow-up were 1.63 (95% CI 1.36–1.95) for the NNRTI versus PI strategies, and 1.12 (0.96–1.31) for the three-class versus combined two-class strategies. The conditional HRs for viral rebound after initial suppression were 0.78 (95% CI 0.64–0.95) for the NNRTI versus PI strategies and 1.05 (0.88–1.24) for the three-class versus two-class strategies (figure 7). Of 1360 participants with an HIV RNA measurement at or after 4 months, 866 (64%) had virological failure: HRs were 0.66 (95% CI 0.56–0.78) for the NNRTI versus PI strategies and 0.87 (0.75–1.0) for the three-class versus combined two-class strategies.

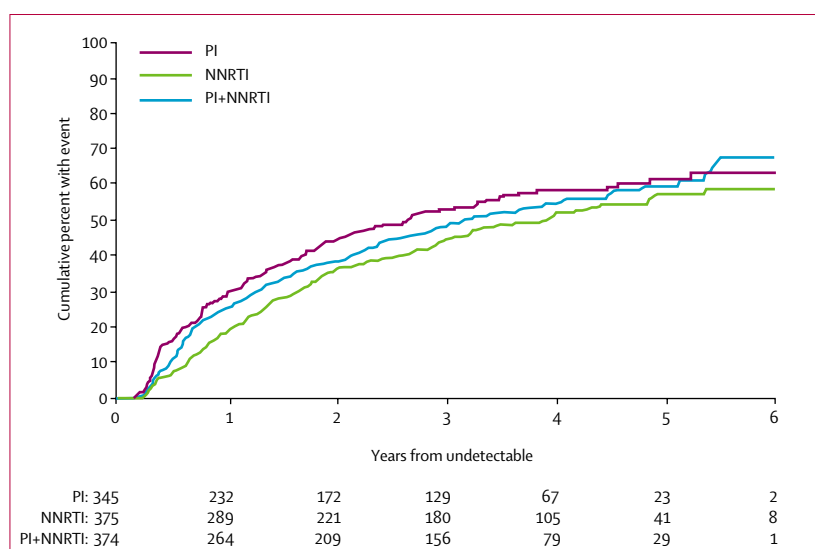


Figure 7: Kaplan-Meier estimates of time to first increase in HIV RNA concentration to more than 400 copies per mL for those who had achieved an HIV RNA concentration of less than 50 copies per mL, by treatment strategy



Virological failure associated with any drug resistance was different for the two-class strategies (HR 0.78 for the NNRTI vs PI comparison, 95% CI 0.61–0.99). Among those with virological failure, multi-class resistance was present in 12%, 19%, and 10% of those in the PI, NNRTI, and three-class strategies, respectively; no resistance was found in 59%, 52%, and 53%. Of those who had no drug resistance, 36%, 48%, and 45% of the PI, NNRTI, and three-class strategies were not on antiretroviral treatment at the time of virological failure.

In the protease gene, the most common mutations identified were Asp30Asn (in 41% of the 74 participants with a protease mutation), Leu33Ile/Phe (in 24%), and Leu90Met (in 24%). In the reverse transcriptase gene, the most common NNRTI mutations were Lys103Asn (in 61% of the 252 participants with a NNRTI mutation), Tyr181Cys/Ile (in 24%) and Gly190Ala/Ser (in 16%); the most common NRTI mutation was Met184Ile/Val (in 79% of the 260 participants with an NRTI mutation). Of those with NRTI resistance, 92%, 71%, and 82% were resistant to lamivudine only in the PI, NNRTI, and three-class strategies, respectively.

Models were analysed separately for each strategy to predict an AIDS-defining event or death with time-dependent covariates corresponding to virological failure with and without resistance and the ability to achieve an HIV RNA concentration of fewer than 400 copies per mL before the event time. The risk of an AIDS-defining event or death for those with virological failure with no resistance was significantly higher than for those who did not experience virological failure (HR 3.6, 95% CI 1.8–7.0 for the PI strategy; 4.7, 2.4–9.5 for the NNRTI strategy; 2.9, 1.6–5.2 for the three-class strategy). In the NNRTI-containing strategies, but not the PI strategy, the risk of an AIDS-defining event or death was higher for those who had virological failure with any resistance compared with those who did not have virological failure (4.5, 2.3–8.7 for the NNRTI strategy; 2.8, 1.6–4.9 for the three-class strategy; 1.6, 0.8–3.5 for the PI strategy).

Overall, 439 patients had at least one grade 4 adverse event. The rate of grade 4 events did not vary by treatment strategy ( $p=0.98$ ). Rises in results of liver function tests (aspartate transferase and alanine transferase) were the most common grade 4 adverse events reported, and they occurred most often in individuals receiving an NNRTI-containing regimen. Rates of increases in results of liver function tests were 0.7, 1.4, and 1.5 for those assigned the PI, NNRTI, and three-class strategies ( $p=0.03$ ). Other common grade 4 adverse events for PI, NNRTI, and three-class strategies were: anaemia (rates=0.7, 1.0, and 0.8;  $p=0.65$ ), neutropenia (rates=0.8, 0.7, and 0.8;  $p=0.98$ ), and gastrointestinal symptoms (nausea, diarrhoea, vomiting, and other gastrointestinal complaints; rates=1.0, 1.2, and 1.1;  $p=0.79$ ).

## Discussion

The FIRST study has the longest follow-up time (median follow-up of 5 years) of any randomised clinical trial to compare three different initial antiretroviral treatment strategies for long-term clinical (AIDS-defining event or death) and immunological outcomes. The findings show that there is no difference between a PI and an NNRTI strategy in the rate of an AIDS-defining event or death or CD4 cell count decline to less than 200 cells per mm<sup>3</sup>. Furthermore, a three-class strategy did not result in a greater increase in CD4 cell count than did a two-class strategy, and was associated with more toxic effects. Our PI versus NNRTI findings are consistent with the findings of the EuroSIDA observational cohort,<sup>3</sup> showing that the rates of clinical outcome of AIDS or death are similar for given CD4 cell counts and HIV RNA concentrations irrespective of antiretroviral treatment regimen used.

The results of the FIRST trial extend the findings of both the ACTG 384<sup>4</sup> and INITIO<sup>5</sup> clinical trials. All three trials showed similar increases in CD4 cell counts during follow-up in all of the study groups and greatest suppression of HIV RNA in the NNRTI-based regimen. In the FIRST trial, time to viral rebound (HIV RNA >400 copies per mL) was also faster for the PI strategy than the NNRTI strategy, a finding similar to published data from the EuroSIDA group.<sup>14</sup> The FIRST study was the only one of the three randomised clinical trials to show a significantly higher rate of drug discontinuation due to toxic effects in the three-class regimen than in the two-class strategies, although INITIO<sup>5</sup> reported similar but not significant findings.

That detection of differences in the extent of suppression of HIV RNA (typically after a year of follow-up) is the best way to ascertain efficacy of one drug (or regimen) versus another has been a central tenet in antiretroviral clinical trials.<sup>15</sup> However, as the results from the FIRST trial show, even consistent differences over time in viral suppression rates favouring those who started an NNRTI strategy compared with those who initiated a PI strategy did not result in differences in the composite endpoint of an AIDS-defining event or death, or in mean change in CD4 cell count. Discordant CD4 cell count and HIV RNA responses have been seen in antiretroviral-naïve patients in other randomised clinical trials. For example, in the lopinavir–ritonavir versus nelfinavir trial, reported by Walmsley and co-workers,<sup>1</sup> there were no differences in CD4 cell count increases at 1 year of follow-up, despite significant differences in HIV RNA concentrations favouring the lopinavir–ritonavir group. Conversely, the Atlantic study<sup>16</sup> reported that a nevirapine-based regimen was associated with a lower increase in CD4 cell count than was an indinavir-based regimen, despite similar HIV RNA responses between the two groups at 2 years. Some studies have suggested that the immunological response to a PI-containing regimen may be greater than to an NNRTI-containing regimen.<sup>16,17</sup> Other studies, like ours, have suggested that the immunological response to

a PI-containing regimen and an NNRTI-containing regimen are similar.<sup>18,19</sup> Why the greater suppression of HIV RNA in the NNRTI and three-class strategies than in the PI strategy in the FIRST trial did not result in a greater increase in CD4 cells over time is not clear. Possibly, low-level viraemia might not immediately affect CD4 cell counts. Raffanti and colleagues,<sup>20</sup> in a retrospective study of 3000 people on antiretroviral treatment, showed that those with HIV RNA concentrations between 400 copies per mL and 20 000 copies per mL had preserved immunological status (CD4 cell counts) compared with those with HIV RNA concentrations greater than 20 000 copies per mL during at least the first year of follow-up. In a more treatment-experienced population, Ledergerber and colleagues<sup>21</sup> showed that suppression of HIV RNA below 10 000 copies per mL or to at least 1.5 log<sub>10</sub> below off-treatment values was not associated with a decline in CD4 cell count. Differences in apoptosis by strategy are not likely to explain these results. Although PIs have been shown to effectively reduce apoptosis even in the absence of complete viral suppression,<sup>22</sup> recent data have shown no difference in the extent of reduction of apoptosis between PI-containing and NNRTI-containing antiretroviral regimens.<sup>23</sup>

Another possible explanation for the lack of a better CD4 cell count response in association with the better viral suppression observed in the NNRTI-containing regimens could be related to the increased risk of an AIDS-defining event or death in those assigned to the NNRTI-containing strategies who developed resistance. In other words, the potential benefits of a more HIV RNA suppressive strategy might be offset by the cost of NNRTI-selected mutations, especially in those individuals with less than perfect adherence. The association of particular drug resistance by class with an AIDS-defining event or death and the association with drug resistance and self-reported adherence to antiretroviral treatment is being further explored with the FIRST data.<sup>24</sup>

In summary, the results of the FIRST trial have shown that the three-class regimen offers no advantages over a two-class strategy for immunological or clinical outcomes and has the disadvantage of increased toxic effects. The FIRST study also showed that initiation of antiretroviral treatment with a two-class NNRTI-based regimen or a two-class PI-based regimen resulted in similar clinical and immunological outcomes after a median of 5 years of follow-up. The FIRST trial was not designed to compare a ritonavir-boosted PI-based regimen with an unboosted PI-based regimen. Hence, it is not known whether initiation of antiretroviral treatment with a ritonavir-boosted PI regimen would show any advantage in the long term over initiation of treatment with an NNRTI-based regimen. However, around 25% of individuals starting treatment that included a PI in the FIRST study did so with a ritonavir-boosted regimen, and many of those who

started on a single PI subsequently switched to a ritonavir-boosted PI by the end of the trial. Therefore, it seems reasonable to conclude that starting treatment with either an NNRTI-based regimen or a PI-based regimen, but not both together, are good strategies for long-term antiretroviral management in treatment-naïve patients with HIV with a wide range of baseline CD4 cell counts and diverse demographics.

#### Contributors

RDM, RMN, GP, LC, MBW, BS, and MD designed the study and wrote the study protocol. RDM, RMN, GP, LC, YX, MJK, MBW, CH, and BS implemented the study. GP, LC, YX, and KHH did the statistical analyses. RDM, RMN, MJK, and MBW were responsible for patient accrual. All authors assisted with interpretation of results and manuscript preparation.

#### Conflict of interest statement

RDM has received speaking honoraria and consulting fees from Bristol-Myers Squibb, GlaxoSmithKline, Abbott, Pfizer, Gilead Sciences, Boehringer-Ingelheim, and Roche and has received speaking honoraria from Merck. RDM has also received grant support from GlaxoSmithKline, Pfizer, and Abbott. RMN has consulting agreements with Abbott, Pfizer, Gilead Sciences, Tibotec, and Boehringer-Ingelheim and also has research contracts with Bristol-Myers Squibb, Abbott, GlaxoSmithKline, and Merck. MJK has received royalties from a patent owned by Stanford University for some HIV genotype resistance tests. MJK is the local primary investigator on two Merck HIV-1 integrase inhibitor studies. In the past, MJK has received honoraria and speakers fees, including reimbursement of travel and accommodation expenses from Abbott, Agouron, Bristol-Myers Squibb, GlaxoSmithKline, and Gilead Sciences. MBW has received speaking honoraria from GlaxoSmithKline, Gilead Sciences, and Roche and has received grant support from Abbott. CH has received honoraria from the CPCRA and has represented the consumer's interests as a member of the FIRST protocol team since its early development. All other authors declare no conflict of interest.

#### Acknowledgments

We thank the people who participated in the study; the staff at participating CPCRA units; Lynn Besch for her many years of involvement on the protocol team; Doug Mayers as former protocol co-chair; David Cohn, Wafaa El-Sadr, Fred Gordin, and Jim Neaton for their thoughtful comments and manuscript review; and Bristol-Myers Squibb and GlaxoSmithKline for provision of some study medication. This study was supported by grants (5U01AI042170-10 and 5U01AI046362-03) from the National Institute of Allergy and Infectious Diseases, National Institutes of Health.

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